NATHAN LOWREY

Electrical Engineering Student | Licensed Apprentice Electrician | Reaud Honors College

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Texas Apprentice Electrician License #831394 | Active through June 2027

EDUCATION

Lamar University - Reaud Honors College | Beaumont, TX *Expected May 2030*

Bachelor of Science in Electrical Engineering

Engineering Organizations: IEEE | AIAA | Engineers Without Borders (EWB)
Engineering Teams: IEEE Robotics Team | Robotic Cardinals

ELECTRICAL EXPERIENCE

IZCO Electrical Services | Apprentice Electrician *June 2026 - August 2026*

- Performed hands-on electrical work across new construction and existing residential systems, including installation and troubleshooting.
- Installed and worked with 120V and 240V systems, including panels, breakers, disconnects, junction boxes, receptacles, switches, and branch-circuit wiring.
- Pulled, routed, and terminated conductors; mounted and terminated electrical panels; and completed device and box installations.
- Bent and installed EMT using hand benders and worked with both EMT and PVC conduit using common electrical, hand, and power tools.
- Worked alongside experienced electricians to troubleshoot installation challenges and solve field problems, developing practical insight into how electrical systems and engineering concepts translate into real-world applications.

ENGINEERING PROJECT EXPERIENCE

Engineers Without Borders (EWB) | Rotary Subwoofer Design & Build | Team Member *August 2026 - Present*

- Collaborate with an engineering team to design, fabricate, wire, and test a custom rotary subwoofer system.
- Contribute to electrical and mechanical integration, including system wiring, structural fabrication, motor mounting, and driveshaft assembly.
- Participate in iterative testing and technical problem-solving, operating system controls and contributing solutions to design and performance challenges.

ENGINEERING & LEADERSHIP EXPERIENCE

NASA High School Aerospace Scholars (HAS) | Senior Mentor *Oct. 2024 - Apr. 2025*

- Selected to return as a Senior Mentor after completing NASA HAS, mentoring 8 students through engineering activities, design challenges, collaborative problem-solving, and technical communication.

NASA Moonshot Engineering Design Challenge | Lead Design Manager *Summer 2024*

- Selected as Lead Design Manager for a 9-member engineering team, coordinating design responsibilities and leading development of a conceptual lunar vehicle for astronaut transportation and sample collection.
- Learned Autodesk Inventor within a five-day design challenge to develop the team's final CAD model, contributing to a 4th-place finish among 24 teams.

COMMUNITY INVOLVEMENT

Church Volunteer Teaching Assistant | K-6 Youth Ministry *2022 - 2026*

- Supported and mentored K-6 students through weekly lessons, developing communication, patience, leadership, and the ability to work effectively with younger students.

TECHNICAL SKILLS

Electrical & Field: Residential Electrical Systems | Conductor Installation & Termination | Panels & Electrical Devices | EMT & PVC Conduit | Hand Conduit Bending | Hand & Power Tools
Engineering Software: MATLAB (Coursework) | AutoCAD (Developing) | Autodesk Inventor (Basic)
Programming: C++ (Basic) | JavaScript (Basic) | Python (Basic)